Problem L. Partially Free Meal

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	512 megabytes

A new restaurant is opened in Byteland. To attract more customers, the meal is partially free. Specifically, there are n types of dishes on sale, labeled by $1, 2, \ldots, n$. Each dish can not be ordered more than once. For the i-th dish, its basic price is a_i dollars, and its event price is b_i dollars. Assume you have ordered k dishes $p_1, p_2, \ldots, p_k$ $(1 \le p_i \le n, p_i < p_{i+1})$, the total amount of dollars that you need to pay for is:

$$\sum_{i=1}^{k} a_{p_i} + \max_{i=1}^{k} \{b_{p_i}\}$$

You are a customer at this restaurant, you decide to order exactly k dishes, what's the minimum possible amount of dollars that you need to pay for?

Input

The first line of the input contains a single integer n $(1 \le n \le 200\,000)$, denoting the number of dishes.

Each of the following n lines contains two integers a_i and b_i $(1 \le a_i, b_i \le 10^9)$, denoting the basic price and the event price of each dish.

Output

Print n lines, the k-th $(1 \le k \le n)$ of which containing an integer, denoting the minimum possible amount of dollars that you need to pay for when you order exactly k dishes.

Example

standard input	standard output
3 2 5 4 3 3 7	7 11 16